EviBound: Evidence-Bound Runtime Enforcement for Smart-Home Agents

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Research Contract
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I. Research Question and Hypothesis

Does validating observation provenance and freshness at execution time reduce unsafe effects beyond an otherwise identical stateful action guard? EviBound binds action execution to validated observation records, rather than planner explanations. We hypothesize a lower unsafe-effect rate than that guard, with benign completion no more than five percentage points below the unprotected agent. These are evaluation targets, not established results. BALD motivates containing unsafe decisions; this study does not reproduce or establish robustness against trained backdoors [1].

II. Scope and Mechanism

To isolate observation integrity from planning complexity, we propose a controlled testbed with three device types: a light and fan (off/low/high), and a timestamped presence sensor. This Project 1 scope requires instructor approval. One local 7B instruction model executes tasks involving one or two actuators and one to three writes; queries are counted separately. An independently built discrete-event state machine is the required backend; SimuHome is optional [2]. Every device write passes through the guard. It enforces device permissions and six state/sequence constraints, including conditional lighting, normalized load limits and intermediate-state checks. It validates referenced observation records against a trusted store, requiring fresh evidence or blocking. Training, physical deployment, full benchmarks and four-system reproduction are excluded.

III. Threat Model and Trust Boundary

A sandbox adversary may alter designated tool-text fields or replay authentic old observations, but cannot directly write devices or alter user authorization, policies, the collector, clock or original record metadata. Protected assets are authorized device states and declared safety invariants. Safety-relevant values are resolved from records, not agent explanations. All conditions share permitted observation access; guards never read hidden labels or future states. External changes occur at declared event boundaries, not inside an atomic check-and-commit step; freshness alone is not a guarantee of continuous-world validity.

IV. Baselines and Experimental Design

Compare B0: unprotected; B1: safety prompt; B2: stateful guard with identical authorization/state rules but no provenance/freshness validation; M: EviBound (B2 plus observation validation). Literature [3]–[9] informs the design, not claims of inherent framework blind spots. Target 24 tasks, each with clean, perturbed and legal-boundary versions, across four conditions and three runs (864 episodes). Assign tool-text interference or replay by task group. Fix model, schemas, resets, event schedules and call/token budgets; allow at most one rejection-triggered re-read/resubmission. Required ablation: remove freshness checks only. A small always-refresh diagnostic tests a simpler explanation. Deterministic component tests are reported separately from model-driven results.

V. Outcomes and Validation

Primary outcome: episodes with an actually executed unsafe state/transition, including intermediate states. Separately report authorization violations, attack-goal success, benign and safe task completion, false blocks on independently legal/evidence-sufficient proposals, re-reads, latency and cost. A separately implemented trajectory oracle scores recorded states; students validate labels and obtain peer/instructor review of rules and representative traces. Retain failed/blocked episodes; report task-paired differences and task-clustered 95% intervals, including null or inconclusive results. Development, pilot and test tasks are disjoint; simulated time is separate from wall-clock latency.

VI. Schedule and Feasibility

Week 4 (current): verify a resettable baseline, automated scoring, $\geq 70\%$ benign completion on a fixed development set, and one repeatable action-changing perturbation. If unmet, report the shortfall and pivot to 12 tasks/one perturbation; any extension needs approval. **Weeks 5–7:** guard, logging and invariant tests. **Week 8:** pilot and metric/protocol freeze. **Weeks 9–10:** freeze tasks/labels before main runs. **Weeks 11–12:** ablation, failure analysis and code freeze. **Weeks 13–15:** independent reproduction, report and demonstration.

VII. Resources, Safety, and Deliverables

Use two available RTX 4090s, one primary model and fixed inference budgets; no paid APIs without approval. Synthetic data and simulated devices only; no real home accounts. Review licenses and sanitize releases. Disclose AI-assisted test-driven development/review; Jev is optional and development-only. Humans approve policies and evidence. Deliver a five-minute demo, six-to-eight-page report, versioned scenarios/prompts, raw traces, analysis and a one-command reproducible subset with a safety statement.

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